

Q7. Logistic Regression — Student Placement Prediction

```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
```

```
# Student Placement Dataset
```

```
X = [
    [9.2, 90, 88],
    [8.8, 85, 84],
    [8.5, 82, 80],
    [7.2, 72, 70],
    [6.8, 68, 65],
    [9.0, 91, 89],
    [7.0, 70, 68],
    [8.7, 86, 83],
    [6.5, 65, 60],
    [9.1, 92, 90]
]
```

```
# 1 = Placed, 0 = Not Placed
```

```
y = [1, 1, 1, 0, 0, 1, 0, 1, 0, 1]
```

```
# Split the dataset
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)
```

```
# Create and train Logistic Regression model
```

```
model = LogisticRegression()
model.fit(X_train, y_train)

# Test accuracy
accuracy = model.score(X_test, y_test)

print("LOGISTIC REGRESSION")
print("-----")
print("Test Accuracy:", round(accuracy * 100, 2), "%")

# New student
new_student = [[8.9, 88, 85]]

prediction = model.predict(new_student)

print("CGPA:", 8.9)
print("Aptitude Score:", 88)
print("Communication Score:", 85)

if prediction[0] == 1:
    print("Predicted Result: PLACED")
else:
    print("Predicted Result: NOT PLACED")
```

OUTPUT

```
Python 3.14.6 (tags/v3.14.6:c63aec6, Jun 10 2026, 10:26:10) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
```

```
= RESTART: C:/Users/yuvas/Downloads/win-g2010-1_3-n_mcd/win/LIB/machine learning lab/EXP 7.py
```

```
LOGISTIC REGRESSION
```

```
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```

```
Test Accuracy: 100.0 %
```

```
CGPA: 8.9
```

```
Aptitude Score: 88
```

```
Communication Score: 85
```

```
Predicted Result: PLACED
```

```
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```